

OPSC6312 PART 1: POE

KUN'WE TYRONE MDAKA: ST10262122
ORATILE MAUNGWA: ST10443081
GONTSE RAKOSA: ST10449265
RICHARD SEBOLA: ST10441486

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Section A: Research Report

Introduction

EduConnect: A Comparative Analysis of Existing Study Applications

Although mobile learning has revolutionised education worldwide (Pew Research Center, 2021), South African students continue to encounter major obstacles such as uneven access to resources, linguistic variety, and irregular internet connectivity (Mkhize & Ndimande-Hlongwa, 2014). By offering a full study software that enables students to create custom quizzes, arrange content by subject, monitor performance, and visualise progress through graphical data, EduConnect seeks to tackle these challenges. We acknowledged that local educational circumstances are frequently overlooked by current implementations (Anderson, 2016). In order to guarantee accessibility and engagement, EduConnect is based on active recall and spaced repetition (Karpicke & Roediger, 2008; Cepeda et al., 2006). It also includes multilingual support, offline capabilities, and real-time notifications (Department of Basic Education, 2020).

This paper examines the features, strengths, shortcomings, and implementation tactics of three Android applications, Quizlet, Kahoot!, and Socratic by Google, in order to inform our design. In order to create an app that specifically addresses the needs of South African learners, we hope to correct their flaws while incorporating their greatest features through this comparative analysis.

Research on existing applications

Quizlet

Overview

Over 300 million students rely on Quizlet, a comprehensive study aid, to help them prepare for exams, making it one of the most popular learning platforms in the world (PCMag, 2025). With the application, users may make study sets, digital flashcards, and practice exams in almost any subject. Active recall and spaced repetition are the main features of Quizlet, which allows students to sort cards into "Know" and "Still learning" categories to monitor their progress. The Google Play Store (2026) states that the app provides a variety of study options, such as flashcards, multiple-choice questions, true/false questions, written answers, and exam-like practice exams.

The platform's development from a basic flashcard tool to an all-encompassing learning ecosystem is indicative of more general developments in educational technology. Digital flashcard apps have shown considerable efficacy in vocabulary acquisition and factual knowledge retention, according to Chen et al. (2018). While the free edition is still useful for basic study needs, premium features include AI-enhanced learning tools, textbook solutions, and ad-free experiences. Additionally, the software lets users store decks for offline use, which makes it useful for students who don't always have internet connection.

Strengths:

- **Versatility and large library:** Users can frequently locate pre-existing study resources for their courses thanks to the millions of flashcard sets made by educators and students worldwide (ElearningWorld, 2025).

- **Adaptive Learning:** According to Dunlosky et al. (2013), adaptive learning technology creates a customised learning experience that targets individual shortcomings by modifying questions based on the user's knowledge and what needs more work.
- **Variety of question types:** The diverse modes, which range from multiple-choice to written responses, support different learning styles and enhance comprehension.
- **Offline access:** Regardless of internet connectivity, users can download study sets for offline usage, guaranteeing continuous learning (Warschauer & Matuchniak, 2010).
- **Progress tracking:** Clear visual feedback on mastery levels is provided by the "Know/Still learning" sorting method.

Weaknesses:

- **Paywall for vital features:** A lot of customers have complained about having to pay for the test option and other useful study options (PCMag, 2025).
- **Problems with image quality:** According to users, flashcards' connected images look small and cannot be enlarged (Google Play Store, 2026).
- **Usability issues:** Study sets may be permanently lost due to the inadvertent swipe-to-delete function (Google Play Store, 2026).
- **Variability in content quality:** User-generated content isn't always precise or thorough (Crompton et al., 2017).
- **Limited gamification:** Although Quizlet has certain game-like features (matching games, for example), it is not as interesting as its rivals.

Implementation Analysis

Quizlet seems to be developed utilising a solid client-server architecture, based on my understanding of Android Studio development. In order to facilitate smooth transitions between mobile and web platforms, the app probably makes use of a RESTful API to synchronise user data between devices (Android Developers, 2024). The offline feature recommends using local storage systems that store flashcard sets for offline access, including RoomDB or SQLite (Android Developers, 2024). Adaptive questioning and Magic Notes are examples of AI-powered features that show interaction with cloud-hosted machine learning models (Mühl Göuser, 2022). Passwords are encrypted using industry-standard hashing algorithms, and user authentication is probably handled by OAuth or comparable SSO protocols. In accordance with current data protection laws, the app's data safety procedures include encrypting data while it's in transit and allowing users to request data deletion (Voigt & Von dem Bussche, 2017).

Kahoot!

Overview

A game-based learning software called Kahoot! turns conventional tests into fun, competitive games (PCMag, 2025). Kahoot! is an iOS and Android app that lets users make their own tests, or "kahoots," and host them live so others can take part in real-time. The software serves students, instructors, and lifelong learners in a variety of educational contexts, including corporate training settings and

classrooms. Users can compete with friends in study leagues, participate in live games organised by others, and finish self-paced tasks.

Growing awareness of the motivational advantages of game-based learning is reflected in Kahoot!'s gamification strategy (Deterding et al., 2011; Kapp, 2012). Students can now simulate tests and review their scores without an internet connection thanks to recent improvements that include test and offline modes. English, Spanish, French, German, Italian, Brazilian Portuguese, and Norwegian are among the languages that the app supports (Google Play Store, 2026).

Strengths:

- **Gamification approach:** Using timers, leaderboards, and vibrant graphics makes learning fun and boosts student participation (Hamari et al., 2016).
- **Real-time competition:** The participatory nature of the live quiz format inspires students and builds a sense of community.
- **Huge content library:** The platform saves customers time when creating material by providing millions of ready-to-play tests on almost any subject.
- **Flexibility:** Both classroom and individual learning environments can be accommodated by offering live tests or self-paced assignments.
- **Analytics and reporting:** To evaluate learning outcomes and pinpoint areas for development, the app offers comprehensive reports (Ebner & Holzinger, 2007).

Weaknesses

- **Time-consuming quiz creation:** Making quizzes by hand can be difficult, particularly for teachers who are preparing several tests (PCMag, 2025).
- **Limited effectiveness for individual study:** Because the app's design places a strong emphasis on social competition, it is less useful for learning alone than in group or classroom settings.
- **Paid features:** A membership is needed for a number of additional capabilities, such as access to an image library and sophisticated question kinds (Google Play Store, 2026).
- **Technical difficulties:** During live games, some users have complained about periodic network problems and registration difficulties.
- **Content repetition:** After extended use, the discovery page's sparse content has been identified as repetitive.

Implementation Analysis

A real-time communication architecture is probably used in Kahoot!'s implementation to provide multiplayer participation and live quiz hosting. To enable instantaneous question transmission and answer submission during live games, WebSocket or other persistent connections would be necessary (Lubbers & Greco, 2010). When the device reconnects to the internet, the app's offline mode recommends using RoomDB or SQLite for local data persistence with synchronisation capabilities. A dynamic content generating system that enables users to add pictures, videos, and different kinds of questions is indicated by the quiz creation functionality. Secure authentication mechanisms are probably used to manage user accounts, and data is encrypted while it's in transit and at rest. A

backend database would be necessary to store and process user performance data for the reporting and analytics functions (Meier, 2012).

Socratic by Google:

Overview:

Google's Socratic is a free AI-powered learning tool that helps college and high school students comprehend their assignments (ToolChase, 2026). The software finds the best internet resources to clarify subjects and lets users ask enquiries with their voice or camera. Algebra, geometry, trigonometry, biology, chemistry, physics, history, and literature are just a few of the many topics covered by Socratic. The app collaborates with educators and professionals to offer visual explanations for every topic, assisting pupils in understanding the ideas behind any issue.

Socratic does not prioritise gamification or quiz production, in contrast to Quizlet and Kahoot!. Rather, it functions as a tool for explaining concepts and helping with homework, offering carefully chosen instructional materials and step-by-step solutions. This strategy is in line with current views on AI in education, which highlight how intelligent tutoring systems may offer individualised assistance (Holmes et al., 2019).

Strengths

- Completely free: Socratic is entirely free; all students, regardless of finances, can use it because it doesn't have any premium tiers, paid plans, or memberships (ToolChase, 2026).
- Google Lens integration: Instant step-by-step explanations can be obtained by taking a picture of an issue.
- Wide-ranging subject coverage: The app offers visual explanations for a variety of topics that accommodate various learning preferences.
- AI-powered question recognition: By removing unnecessary material, the app's AI and carefully chosen resources provide students with just what they need.
- Equitable access: In line with inclusive education ideals, the app is available to all students, regardless of their financial situation (Booth & Ainscow, 2011).

Weaknesses

- Quick fixes over guided learning: The app's design prioritises quick fixes over in-depth comprehension, which could promote surface-level engagement (ToolChase, 2026).
- Dependency on online resources: When subjects grow complex or difficult to understand, the application may have trouble.
- No quiz creation: Its lack of quiz authoring and progress tracking restricts its usefulness for performance monitoring and self-evaluation.

- Potential of academic dishonesty: According to Gwinnett Magazine (2022), some professors are concerned that pupils utilise the app to avoid learning rather than complete assignments.
- Prioritise problem-solving over concept mastery; this may not help with long-term recall (Mayer, 2002).

Implementation Analysis

Based on Google's AI infrastructure, Socratic uses machine learning models and the Google Knowledge Graph to identify and respond to queries (Google, 2024). Optical character recognition (OCR) technology, perhaps driven by Google Lens, is used by the app's camera-based question scanning feature to decipher handwritten or printed difficulties. Integration with speech-to-text APIs is indicated by the voice input feature. A web scraping and content curation system that scans and extracts instructional materials from all over the internet is suggested by the app's capacity to locate pertinent online resources. The app does not share user data with outside parties, and user data is encrypted during transmission. Compared to Quizlet or Kahoot!, the app's lightweight design and absence of quiz elements point to a simpler architecture with less focus on user-generated content and data synchronisation.

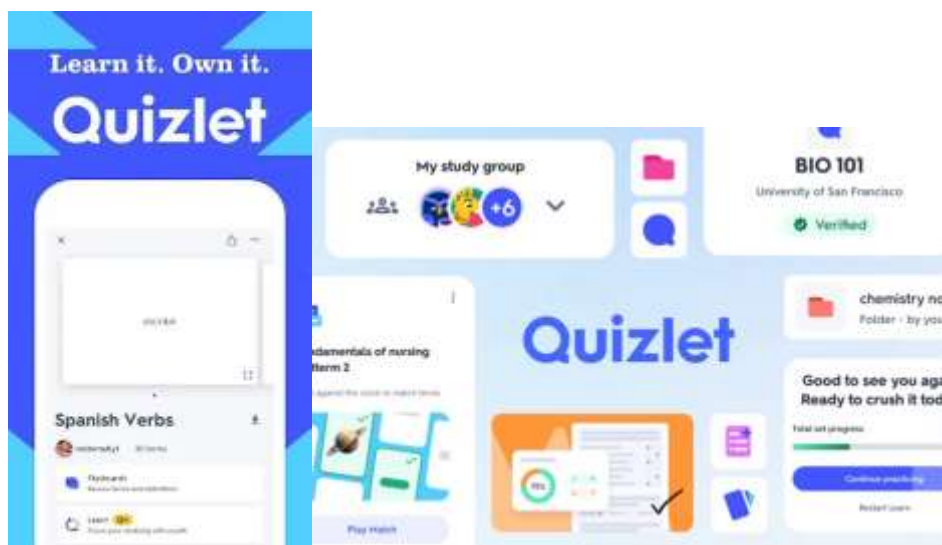
Comparison of all three apps:

Feature	Quizlet	Kahoot!	Socratic
Primary Purpose	Flashcard-based study and test preparation (PCMag, 2025)	Game-based quiz learning (PCMag, 2025)	Homework help and concept explanation (ToolChase, 2026)
Quiz Creation	Yes (flashcards and practice tests) (Google Play Store, 2026)	Yes (custom quizzes with images/videos) (Google Play Store, 2026)	No
Progress Tracking	Yes (Know/Still learning sorting) (ElearningWorld, 2025)	Yes (reports and analytics) (PCMag, 2025)	No
Offline Mode	Yes (save decks for offline access) (Google Play Store, 2026)	Yes (offline mode introduced) (Google Play Store, 2026)	Limited
Gamification	Limited (matching games) (ElearningWorld, 2025)	Extensive (leaderboards, timers, music) (PCMag, 2025)	None

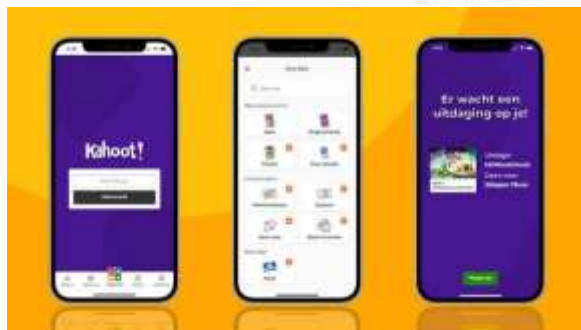
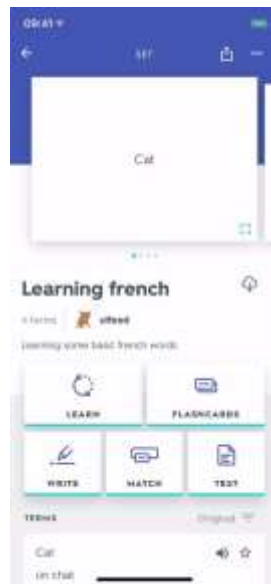
Feature	Quizlet	Kahoot!	Socratic
AI Features	Yes (adaptive learning, Magic Notes) (Laxu AI, 2026)	Limited	Yes (Google AI, Lens integration) (ToolChase, 2026)
Multi-language Support	Yes (Google Play Store, 2026)	Yes (multiple languages) (Google Play Store, 2026)	Yes
Pricing Model	Freemium (subscription required for advanced) (PCMag, 2025)	Freemium (free for teachers, paid for advanced) (PCMag, 2025)	Completely free (ToolChase, 2026)
Content Library	Millions of user-generated sets (ElearningWorld, 2025)	Millions of ready-to-play quizzes (PCMag, 2025)	Curated web resources (ToolChase, 2026)
Best Use Case	Individual study and memorisation	Group/classroom engagement	Quick problem-solving assistance

Screenshots of all three apps

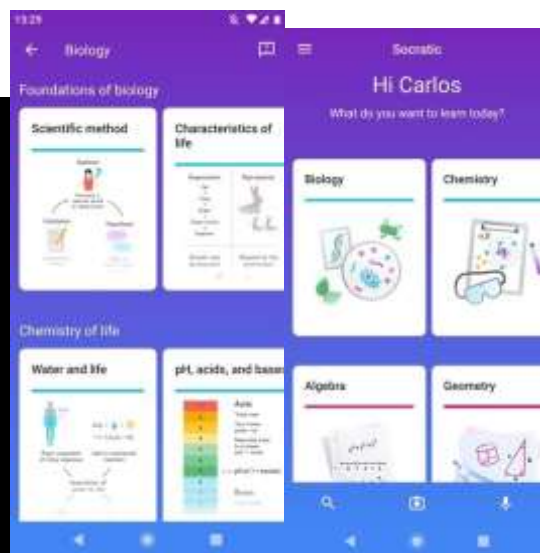
Screenshots of Quizlet:



Screenshots of Kahoot!:



Screenshot of Socratic by Google



Best features to incorporate into EduConnect:

The elements listed below are the most beneficial to include in EduConnect, according to our examination of the three programs. These elements were chosen because they support the app's main goals of giving South African pupils individualised, interesting, and accessible learning opportunities.

From Quizlet:

- **Adaptive Quizzing:** EduConnect's objective of assisting students in identifying areas of difficulty and growth is perfectly aligned with the ability of quizzes to adjust based on what the student knows and what need more study. Adaptive learning techniques that focus on areas of weakness are more successful than consistent practice, according to Karpicke and Roediger (2008). This element, in our opinion, is crucial for delivering individualised learning experiences that cater to each student's needs.
- **Multiple Study Modes:** Different learning styles and cognitive preferences can be accommodated by providing a variety of question formats, such as flashcards, multiple-choice, true/false, and written responses. Diverse instructional approaches are beneficial to students, according to research by Fleming and Baume (2006). We believe that this diversity is essential for engaging students with varying learning preferences.
- **Offline Mode with Synchronisation:** Students with erratic internet access must be able to study offline and synchronise their progress when they reconnect. We think this feature is essential for guaranteeing fair access to learning resources in the South African context, since many pupils still struggle with dependable internet connectivity (Department of Basic Education, 2020).
- **User-Generated Content:** The main goal of EduConnect is to let students make their own tests and arrange them by subject. User-generated material in educational applications encourages participation and ownership of learning, as noted by Collins and Halverson (2018). This functionality is essential to the value proposition of the app, in our opinion.

From Kahoot!:

- Timers, points, and leaderboards are examples of gamification elements that increase motivation and engagement while maintaining student interest (Hamari et al., 2016).
- **Real-Time alerts:** By encouraging regular study habits, push alerts for updates and reminders improve engagement (Wang et al., 2014).
- **Performance Analytics:** Detailed reports enable students to identify areas of weakness and make well-informed choices regarding their academic endeavours.

From Socratic by google:

- **AI-Powered Assistance:** Giving students resources and explanations when they are having difficulties provides individualised support at scale (Holmes et al., 2019).
- **Accessibility:** By removing cost obstacles, a generous or free tier makes the program egalitarian for underprivileged areas (Warschauer & Matuchniak, 2010).
- **Voice and Camera Input:** Multimodal input techniques, such as voice and camera, increase accessibility for users with different levels of technical proficiency and literacy (Jarvis, 2020).

Additional Features for EduConnect:

- Critical language hurdles to educational involvement are addressed by providing support in English, isiZulu, and Afrikaans (Mkhize & Ndimande-Hlongwa, 2014).
- Single Sign-On (SSO): SSO increases user adoption and retention by streamlining login and lowering friction (Anderson, 2016).
- Subject Organization: Students can better manage their learning across several disciplines when content is arranged by subject and topic.
- Graphical Progress Visualisation: By displaying performance patterns, charts and graphs allow for evidence-based learning decisions (Tufte, 2001).
- REST API Integration: Provides a smooth user experience by enabling data synchronisation and persistence between devices.

Conclusion

A comparison of Quizlet, Kahoot!, and Socratic reveals clear advantages that we may take advantage of: free AI-powered support, engaging gamification, and customised adaptive learning. EduConnect will incorporate them while resolving their main flaws, which include inadequate progress visualisation and constrictive paywalls. In order to differentiate ourselves in the educational landscape, we prioritise accessibility and evidence-based learning. By combining these components with our distinctive value proposition, custom quizzes, subject organization, graphical performance tracking, and multilingual support, we hope to create a potent, inclusive study tool for South African students.

SECTION B: PLANNING AND DESIGN DOCUMENT

1. Introduction

EduConnect: A Comprehensive Study Application

EduConnect is a study and quiz application created to help students learn in a more personalised, engaging, and accessible way. The application gives students the ability to create their own quizzes, organise learning materials according to different subjects, monitor their academic performance, and view their progress using graphs and other visual tools. By showing students which topics they understand well and which areas need more attention, EduConnect encourages independent learning and continuous academic improvement.

The application is also designed with the needs of South African students in mind. Many students experience challenges such as language differences, limited digital skills, and unreliable internet access. To help overcome these challenges, EduConnect includes support for multiple languages such as English, isiZulu, and Afrikaans, as well as offline functionality and real-time notifications. These features make the application more accessible to students from different backgrounds and with different levels of access to technology and the internet.

This document presents the overall planning and design of EduConnect. It covers the application's main features, system requirements, user interface, API structure, data models, and development timeline. The application follows a user-centred design approach, with the main goal of providing students with a simple, effective, and enjoyable learning experience.

2. App Overview

Application Name: EduConnect

Icon Design:



The proposed EduConnect icon uses a modern educational design that combines an open book with connected puzzle pieces that form the shape of a graduation cap. The colour scheme transitions from deep blue to vibrant orange, representing knowledge, development, creativity, and energy.

The puzzle pieces represent how different ideas and concepts are connected during the learning process, while the graduation cap represents education, academic progress, and achievement. Together, these elements communicate the main purpose of EduConnect: helping students connect what they learn and work towards their academic goals.

Tagline: *"Connect. Learn. Achieve."*

Innovative Features

EduConnect includes a range of features designed to make studying more interactive and effective:

1. AI-Powered Adaptive Quizzes:

The quiz system adjusts questions according to a student's performance. Students receive more questions on topics they struggle with while continuing to practise areas where they are already performing well.

2. Visual Progress Dashboard:

Students can easily view their academic performance through graphs and visual reports. The dashboard can show performance trends, subject progress, and areas that require additional attention.

3. Subject-Based Content Organisation:

Students can organise their quizzes and study materials by subject, topic, and difficulty level. This makes it easier to find specific content and plan study sessions.

4. Offline Mode and Synchronisation:

Students can access study materials and complete quizzes without an internet connection. Once the device reconnects to the internet, their results and progress are automatically synchronised.

5. Multi-Language Support:

EduConnect supports English, isiZulu, and Afrikaans, making the application more accessible to a wider range of South African students. The application can also detect the preferred language configured on the user's device.

6. Single Sign-On (SSO):

Students can log in quickly using accounts such as Google, Facebook, or supported institutional accounts, reducing the need to remember another separate password.

7. Real-Time Notifications:

Students receive notifications about upcoming quizzes, study reminders, progress updates, and other important learning activities.

8. Gamification:

Points, badges, achievements, and leaderboards are included to make studying more enjoyable and encourage students to remain consistent with their learning.

9. Peer-to-Peer Challenges:

Students can challenge their friends to quizzes and compare their results. This adds a social and competitive element to the learning experience.

10. Educational Resource Library:

The application provides a collection of useful learning resources and explanations to help students understand topics they may find difficult.

3. Detailed Requirements

3.1 Functional Requirements

Functional Requirements for EduConnect

User Authentication and Security

To ensure that student accounts and personal information remain secure, EduConnect will provide several authentication and security features.

High-Priority Requirements

- Users must be able to create an account using their email address and a password.
- All passwords must be securely protected using the BCrypt hashing algorithm.
- The application must support Single Sign-On (SSO) through Google OAuth 2.0, allowing users to log in quickly and securely.
- User sessions must be managed using JSON Web Tokens (JWT) to maintain secure access across the application.

Medium-Priority Requirements

- Users should be able to reset their passwords through an email-based recovery process.

Low-Priority Requirements

- The application should support biometric authentication, such as fingerprint or facial recognition, for an additional layer of security.

User Profile and Settings

EduConnect will allow users to personalise their accounts and adjust application settings according to their preferences.

High-Priority Requirements

- Users must be able to manage their profiles and upload a profile picture or avatar.
- Users should be able to customise notification preferences, including push notifications, sounds, and vibration settings.
- The application must provide language options for English, isiZulu, and Afrikaans.
- Users should be able to update their email address, password, and other security settings whenever necessary.

Medium-Priority Requirements

- Users should be able to choose between light mode, dark mode, or system-default themes.
- The application should include an account deletion feature along with the ability to export personal data before account removal.

Content and Quiz Management

The platform will provide students with tools to create, organise, and manage study content efficiently.

High-Priority Requirements

- Students must be able to create their own quizzes using different question formats, including multiple-choice, true/false, and written-response questions.
- Quizzes and study materials should be organised according to subjects and topics for easier access.
- Students should be able to save study content for offline use.
- Users must have the ability to edit or delete quizzes they have created.

Medium-Priority Requirements

- Students should be able to assign difficulty levels such as easy, medium, or hard to individual quiz questions.

Low-Priority Requirements

- Students should have the option to share quizzes with other users.

Quiz Taking and Adaptive Learning

EduConnect aims to create a personalised learning experience by adapting quizzes to each student's performance.

High-Priority Requirements

- Quiz questions should be presented one at a time to help students focus.
- The application must use adaptive learning techniques to adjust questions based on student performance in real time.
- Students should receive instant feedback after answering a question, including explanations where necessary.
- Users must be able to review completed quizzes and view the correct answers.

Medium-Priority Requirements

- Students should be able to retake quizzes to improve their understanding and scores.
- Timed quizzes should be available for students who want to practise under exam conditions.

Progress Tracking and Analytics

The application will provide detailed insights into student performance and learning progress.

High-Priority Requirements

- Students must be able to monitor key performance indicators such as average scores and the number of completed quizzes.
- Subject-specific performance tracking should be available.
- A visual dashboard with charts and graphs should display learning progress.
- The system should identify weak areas and recommend topics that require more attention.
- Students should be able to view performance trends over time.

Low-Priority Requirements

- Users should have the option to export performance reports in PDF format.

Offline Mode and Synchronisation

Since many students may experience unreliable internet access, EduConnect will support offline learning and data synchronisation.

High-Priority Requirements

- Students must be able to access previously saved quizzes and study resources while offline.
- Quiz attempts completed offline should be stored locally on the device.
- Once an internet connection is restored, progress and data should automatically synchronise with the server.

Medium-Priority Requirements

- The system should handle conflicts that may occur when changes are made offline and later synchronised.
- Users should be able to view the status and progress of synchronisation activities.

Real-Time Notifications

EduConnect will use notifications to keep students informed and engaged with their studies.

High-Priority Requirements

- Students should receive daily study reminders through push notifications.
- Notifications should be sent when quiz results become available.

Medium-Priority Requirements

- Users should receive reminders about upcoming quiz deadlines.
- The system should group and prioritise notifications to reduce clutter.

Low-Priority Requirements

- Students should receive notifications when friends send quiz challenges.

Multi-Language Support

To support a diverse student population, EduConnect will provide multilingual functionality.

High-Priority Requirements

- The application must support English, isiZulu, and Afrikaans.
- The system should automatically detect and apply the device's preferred language settings.
- Users must be able to switch languages manually through the settings menu.

Medium-Priority Requirements

- Content should be displayed using localised formats for dates, times, numbers, and other regional settings.

Gamification and Student Engagement

To encourage consistent studying and increase motivation, EduConnect will include gamification features.

High-Priority Requirements

- Students should earn points for completing quizzes and achieving good results.

Medium-Priority Requirements

- The system should award badges for important milestones, such as completing a first quiz, achieving a perfect score, or finishing multiple quizzes.

Low-Priority Requirements

- Leaderboards should be available to encourage friendly competition among students.
- Students should be able to challenge friends to quizzes and compare their results.

These requirements provide the foundation for EduConnect's functionality and ensure that the application delivers a secure, engaging, and accessible learning experience for students across South Africa.

3.2 Non-Functional Requirements

Non-Functional Requirements

The non-functional requirements describe the standards and quality levels that EduConnect should meet. These requirements focus on areas such as performance, reliability, security, and usability to ensure that the application provides a smooth and dependable experience.

Performance Requirements

EduConnect should respond quickly and use system resources efficiently so that students can use the application without unnecessary delays.

- **Application Launch Time:**
The application should open and become usable within 2 seconds under normal conditions.
Priority: High
- **Scrolling Performance:**
Scrolling through screens and content should remain smooth, targeting approximately 60 frames per second (60fps).
Priority: High
- **Network Request Caching:**
The application should cache frequently used network data to reduce unnecessary requests and aim to lower overall data usage by approximately 30%.
Priority: Medium
- **Image Loading:**
Images should load progressively so that users can begin viewing content without waiting for every image to load completely.
Priority: Medium

Reliability Requirements

The system should remain dependable and continue providing useful functionality even when technical or network problems occur.

- **Backend Availability:**
The backend API should maintain an uptime of at least 99.9% to ensure that students can access the application's online features reliably.
Priority: High
- **Network Error Handling:**
The application should handle connection failures properly and provide a suitable fallback instead of crashing or becoming unusable.
Priority: High
- **Automatic Request Retry:**
When a network request fails temporarily, the application should automatically attempt the request again using an exponential backoff strategy.
Priority: Medium

Security Requirements

Security is important because EduConnect will handle user accounts, personal information, quiz results, and other potentially sensitive data.

- **Local Data Encryption:**
Data stored on the user's device should be protected using secure Android technologies such as the Android Keystore and EncryptedSharedPreferences.
Priority: High
- **Secure Network Communication:**
Communication between the application and backend services should be protected using **TLS 1.3** to prevent unauthorised access to data during transmission.
Priority: High

- **Input Validation and Sanitisation:**

All information entered by users should be validated and sanitised before being processed or stored. This will help prevent invalid data and common security vulnerabilities.

Priority: High

Usability and Accessibility Requirements

EduConnect should be simple to understand and accessible to students with different levels of technical experience and accessibility needs.

- **Accessibility Support:**

The application should support accessibility features such as screen readers, high-contrast content, and adjustable text sizes.

Priority: Medium

- **Easy and Intuitive Navigation:**

The application should use a clear visual structure and consistent navigation so that users can quickly find the features they need.

Priority: High

- **Onboarding and Help:**

New users should receive a short tutorial when they first use the application. Contextual help should also be available where necessary to explain unfamiliar features.

Priority: Medium

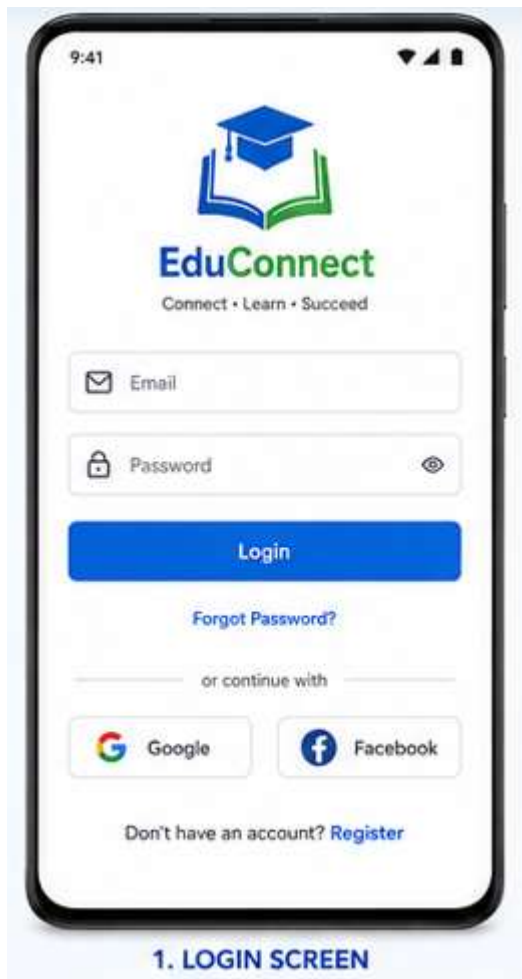
Together, these non-functional requirements will help ensure that EduConnect is not only functional but also fast, secure, reliable, accessible, and easy to use. This will contribute to a better overall experience for students and support the application's goal of providing an effective learning platform.

4. User Interface Design

4.1 Screen Mockups and Descriptions

1. Login Screen

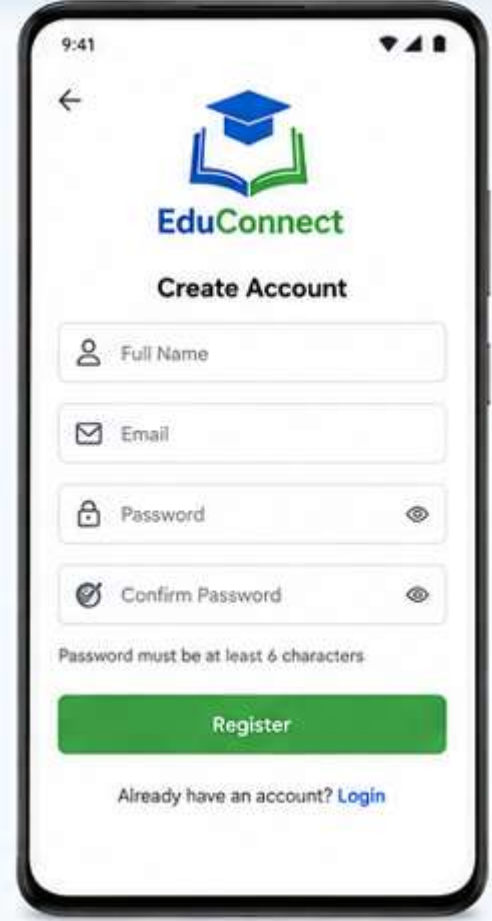
The Login Screen is where existing users access their EduConnect account. It contains an email field, password field, Login button, Forgot Password option, and social login options such as Google and Facebook. New users can select Register to create an account.



1. LOGIN SCREEN

2. Register Screen

The Register Screen allows new students, teachers, or other users to create an EduConnect account. It includes fields for full name, email, password, and confirm password, as well as a Register button. It also provides a link for users who already have an account to return to Login.

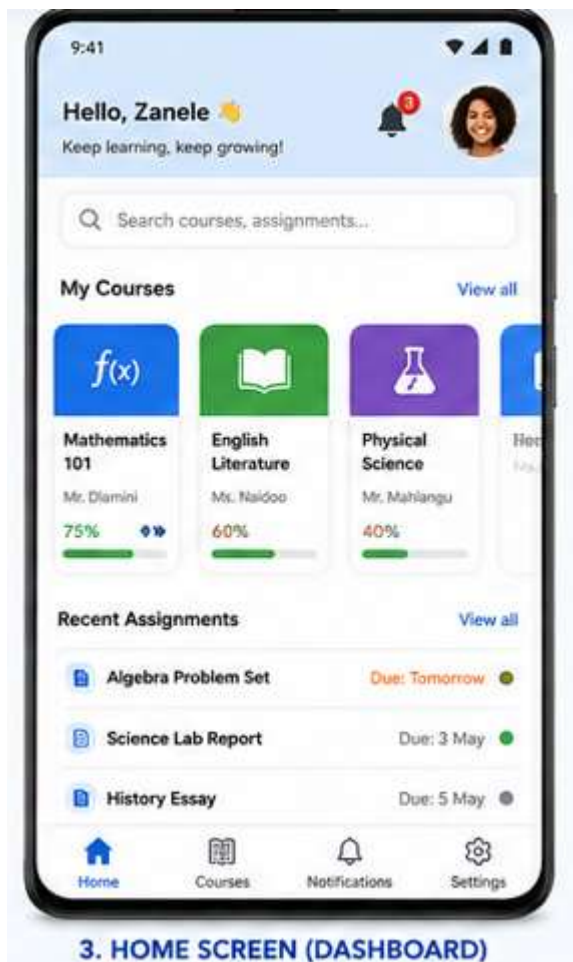


The screenshot shows a mobile application interface for the 'EduConnect' app. At the top, the status bar displays the time '9:41' and signal/battery icons. Below the status bar is a back arrow icon. The app's logo, featuring a blue graduation cap above an open book with green pages, is centered, followed by the text 'EduConnect' in blue and green. Below the logo is the title 'Create Account' in bold. The form consists of four input fields: 'Full Name' (with a person icon), 'Email' (with an envelope icon), 'Password' (with a lock icon and a toggle eye icon), and 'Confirm Password' (with a checkmark icon and a toggle eye icon). Below the 'Confirm Password' field is a text requirement: 'Password must be at least 6 characters'. A large green 'Register' button is positioned below the form. At the bottom, there is a link: 'Already have an account? [Login](#)'. The entire screen is framed by a light blue border.

2. REGISTER SCREEN

3. Home Screen / Dashboard

The Home Screen is the main screen users see after logging in. It provides a quick overview of their learning activities, including My Courses, course progress, recent assignments, notifications, and search. The bottom navigation allows users to quickly access Home, Courses, Notifications, and Settings.



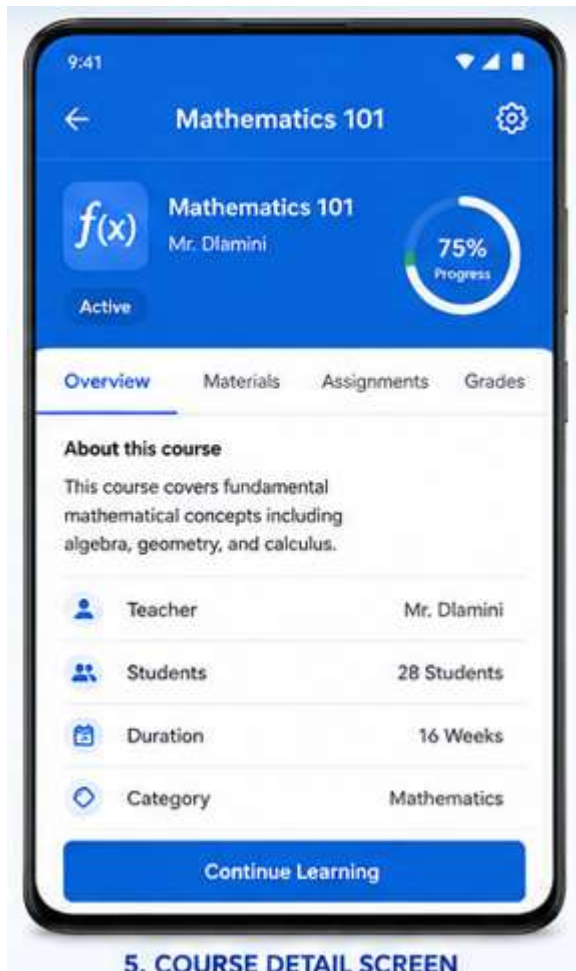
4. Courses Screen

The Courses Screen displays all the courses the user is enrolled in. Users can search for courses and see information such as the course name, teacher, progress percentage, and course icon. Selecting a course takes the user to its detailed course page.



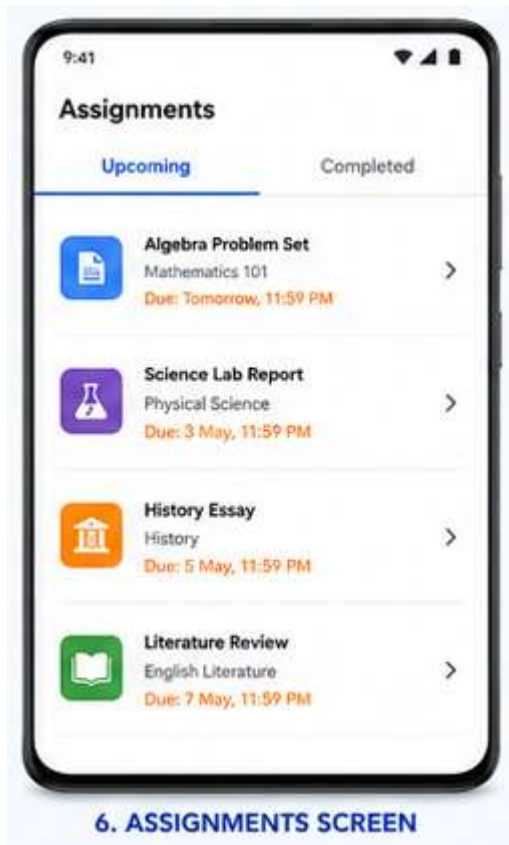
5. Course Detail Screen

The Course Detail Screen provides more information about a selected course, such as Mathematics 101. It shows the teacher, course description, overall progress, and different sections such as Assignments, Grades, and Announcements. Students can use this screen to continue their learning and monitor their performance.



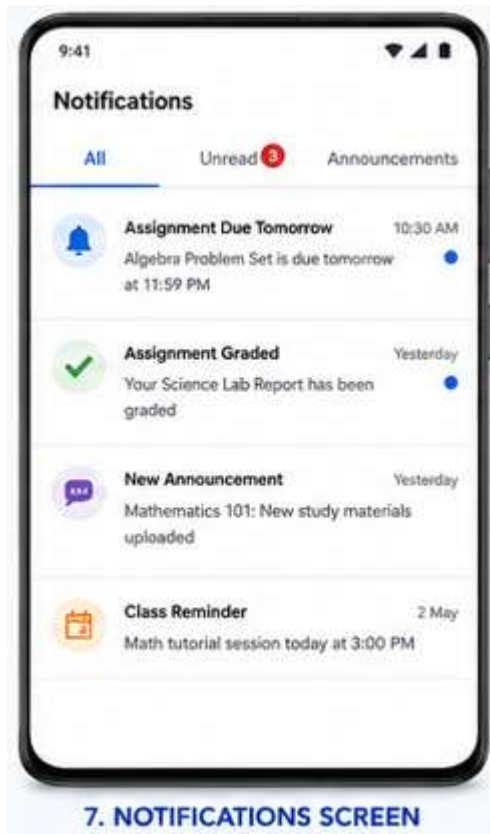
6. Assignments Screen

The Assignments Screen allows students to manage their academic tasks. It separates assignments into upcoming and past assignments and displays information such as the assignment title, course, due date, remaining time, completion status, and grades where available.



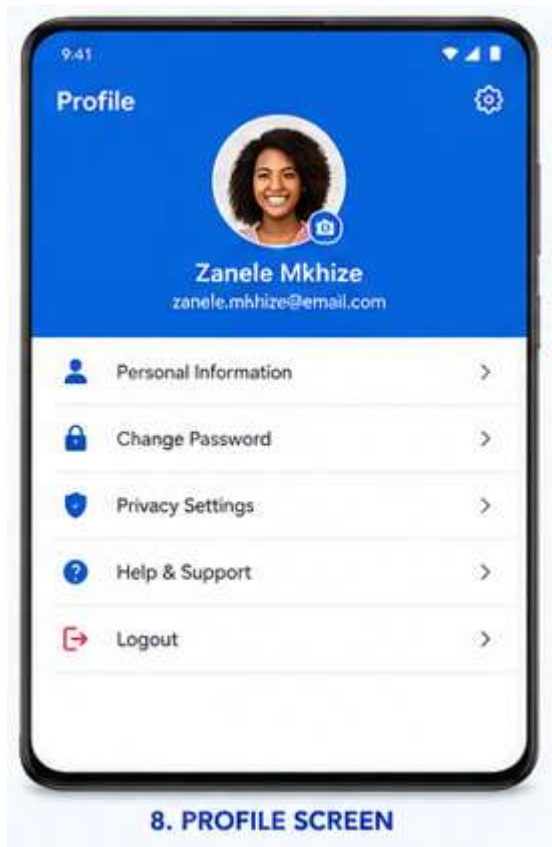
7. Notifications Screen

The Notifications Screen keeps users updated about important academic activities. Notifications can include new assignments, updated grades, course announcements, and assignment reminders. Unread notifications are highlighted so that users can easily identify new information.



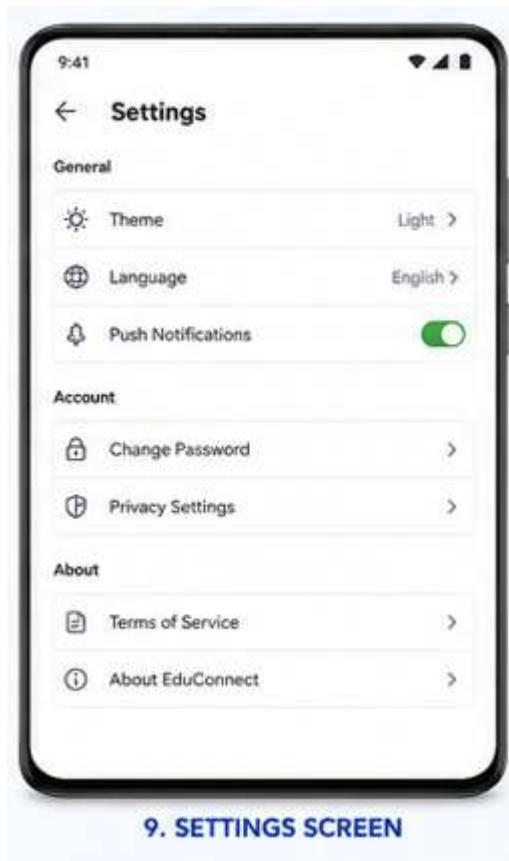
8. Profile Screen

The Profile Screen displays the user's personal account information. In the mock-up, it shows the user's profile picture, name, and email address, along with options such as personal information, changing the password, privacy settings, help and support, and logout.



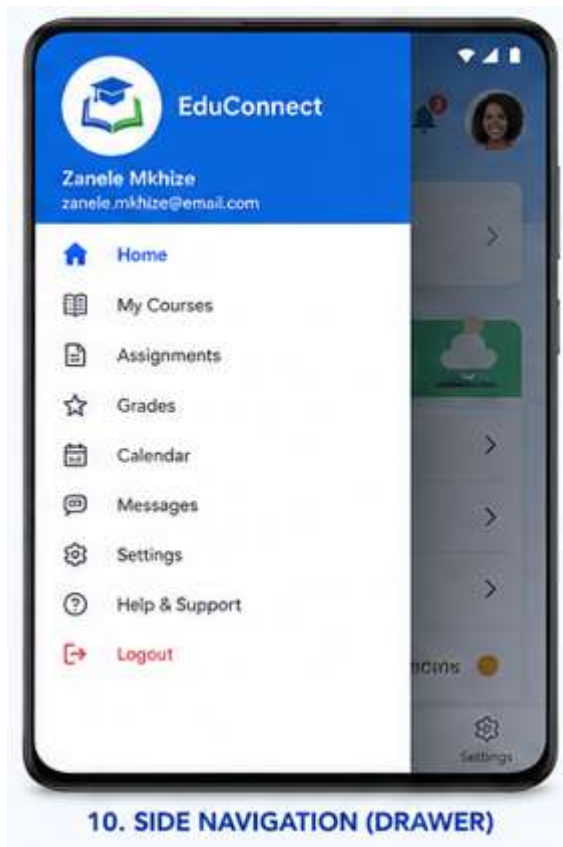
9. Settings Screen

The Settings Screen allows users to customise and manage their EduConnect account. It includes options for theme, language, notifications, password, privacy, terms of service, and information about the app. A logout option is also available. The original design specifies English as the selected language and version 1.0.0.



10. Side Navigation / Navigation Drawer

The Side Navigation Drawer provides another way for users to move around the application. It contains shortcuts to important sections such as Home, My Courses, Assignments, Grades, Calendar, Messages, Settings, Help & Support, and Logout. This makes navigation easier because users can access major features from one menu.



Overall Flow

Login → Home → Courses → Course Detail → Assignments / Grades

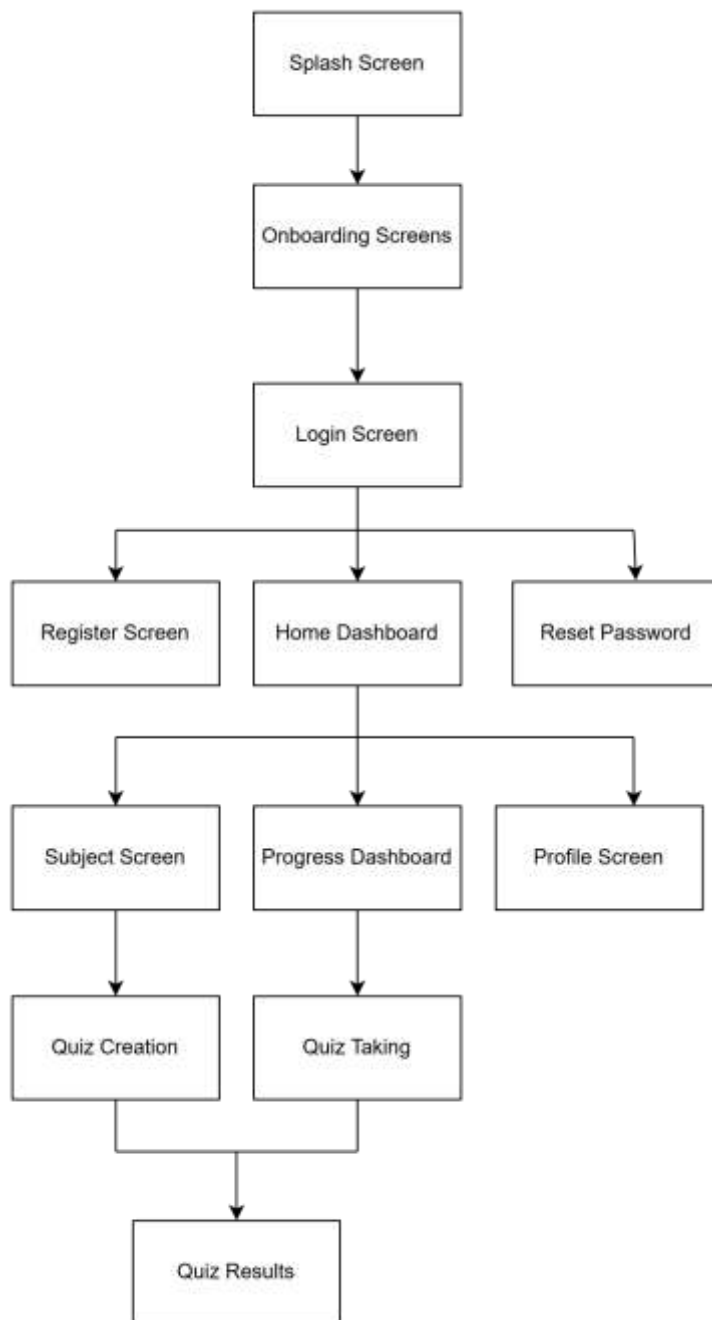
The user can also access:

Home → Notifications

Home → Settings

Settings → Logout → Login

4.2 Navigation Diagram



5. API Design and Architecture

5.1 Backend API Specifications

The MERN stack (MongoDB, Express.js, React for admin panel, Node.js) will be used to build the EduConnect API, which will be hosted on AWS EC2 with an RDS database.

Authentication Endpoints:

Method	Endpoint	Description
--------	----------	-------------

POST	/api/v1/auth/register	Registering a user with a password and email
POST	/api/v1/auth/login	The user logs in using their password and email.
POST	/api/v1/auth/google	Google Single Sign-On verification
POST	/api/v1/auth/refresh	Update the JWT token
POST	/api/v1/auth/forgot-password	Send an email to reset your password
POST	/api/v1/auth/reset-password	Reset password
POST	/api/v1/auth/logout	User logout

User Endpoints:

Method	Endpoint	Description
GET	/api/v1/users/me	Get current user profile
PUT	/api/v1/users/me	Update user profile
PUT	/api/v1/users/me/password	Update password
PUT	/api/v1/users/me/settings	Update user settings
DELETE	/api/v1/users/me	Delete account
GET	/api/v1/users/me/progress	Get user progress data

Quiz Endpoints:

Method	Endpoint	Description
POST	/api/v1/quizzes	Create a quiz
GET	/api/v1/quizzes	List quizzes with pagination
GET	/api/v1/quizzes/:id	Get quiz details

PUT	/api/v1/quizzes/:id	Update quiz
DELETE	/api/v1/quizzes/:id	Delete quiz
POST	/api/v1/quizzes/:id/take	Submit quiz attempt
GET	/api/v1/quizzes/:id/results	Get quiz results

Progress and Analytics Endpoints:

Method	Endpoint	Description
GET	/api/v1/progress/overview	Get overall progress overview
GET	/api/v1/progress/subjects	Get subject-wise progress
GET	/api/v1/progress/trends	Get performance trends over time
GET	/api/v1/progress/improvement	Get areas for improvement
POST	/api/v1/progress/export	Export progress report

Notification Endpoints:

Method	Endpoint	Description
GET	/api/v1/notifications	Get user notifications
PUT	/api/v1/notifications/:id/read	Mark notification as read
PUT	/api/v1/notifications/read-all	Mark all notifications as read
DELETE	/api/v1/notifications/:id	Delete notification

Authentication and security

JSON Web Tokens (JWT) are used by the application for authentication. The following assertions are included in tokens that are signed using HMAC-SHA256:

```
{
  "sub": "user_id",    // Subject - User ID
  "email": "user@email.com",
  "iat": 1692000000,   // Issued at
  "exp": 1692003600,   // Expiration (1 hour from issued at)
  "iss": "educonnect-api", // Issuer
}
```

```
"aud": "educonnect-app" // Audience
}
```

Request/Response Examples

Register User:

Request:

POST /api/v1/auth/register

Headers:

Content-Type: application/json

Body:

```
{
  "name": "John Doe",
  "email": "john@example.com",
  "password": "SecurePass123!",
  "language": "en"
}
```

Response (200 OK):

```
{
  "success": true,
  "data": {
    "user": {
      "id": "usr_123456",
      "name": "John Doe",
      "email": "john@example.com",
      "preferences": {
        "language": "en",
        "theme": "system",
        "notifications": {
          "push": true,
          "email": true,
          "sound": true,

```

```
    "vibration": true
  }
},
"createdAt": "2026-08-15T10:30:00Z"
},
"token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
"refreshToken": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."
}
}
```

Error Response (400 Bad Request):

```
{
  "success": false,
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "Email already exists",
    "details": {
      "field": "email",
      "value": "john@example.com"
    }
  }
}
```

Login User:

Request:

POST /api/v1/auth/login

Headers:

Content-Type: application/json

Body:

```
{
  "email": "john@example.com",
```

```
"password": "SecurePass123!"
}
```

Response (200 OK):

```
{
  "success": true,
  "data": {
    "user": {
      "id": "usr_123456",
      "name": "John Doe",
      "email": "john@example.com",
      "preferences": { ... },
      "progress": {
        "totalQuizzes": 15,
        "averageScore": 78,
        "studyStreak": 5,
        "lastStudyDate": "2026-08-15T09:00:00Z"
      }
    },
    "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
    "refreshToken": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."
  }
}
```

Create Subject:

Request:

POST /api/v1/subjects

Headers:

Authorization: Bearer <token>

Content-Type: application/json

Body:

```
{
```

```
"name": "Chemistry",  
"icon": "🧪",  
"color": "#4CAF50"  
}
```

Response (201 Created):

```
{  
  "success": true,  
  "data": {  
    "subject": {  
      "id": "sub_789012",  
      "userId": "usr_123456",  
      "name": "Chemistry",  
      "icon": "🧪",  
      "color": "#4CAF50",  
      "progress": {  
        "quizzesCompleted": 0,  
        "averageScore": 0,  
        "masteryLevel": 0  
      },  
      "createdAt": "2026-08-15T11:00:00Z",  
      "updatedAt": "2026-08-15T11:00:00Z"  
    }  
  }  
}
```

Create Quiz:

Request:

POST /api/v1/quizzes

Headers:

Authorization: Bearer <token>

Content-Type: application/json

Body:

```
{
  "title": "Chemistry Quiz 1",
  "subjectId": "sub_789012",
  "difficulty": "medium",
  "questions": [
    {
      "type": "multiple-choice",
      "question": "What is the chemical symbol for water?",
      "options": ["H2O", "CO2", "NaCl", "HCl"],
      "correctAnswer": 0,
      "explanation": "Water is composed of two hydrogen atoms and one oxygen atom."
    },
    {
      "type": "true-false",
      "question": "The atomic number of carbon is 6.",
      "options": ["True", "False"],
      "correctAnswer": 0,
      "explanation": "Carbon has 6 protons, so its atomic number is 6."
    }
  ]
}
```

Response (201 Created):

```
{
  "success": true,
  "data": {
    "quiz": {
      "id": "qiz_345678",
      "userId": "usr_123456",
      "subjectId": "sub_789012",
```

```
"title": "Chemistry Quiz 1",
"difficulty": "medium",
"questions": [
  {
    "id": "qst_901234",
    "type": "multiple-choice",
    "question": "What is the chemical symbol for water?",
    "options": ["H2O", "CO2", "NaCl", "HCl"]
  },
  {
    "id": "qst_901235",
    "type": "true-false",
    "question": "The atomic number of carbon is 6.",
    "options": ["True", "False"]
  }
],
"isPublic": false,
"timesTaken": 0,
"averageScore": 0,
"createdAt": "2026-08-15T11:00:00Z",
"updatedAt": "2026-08-15T11:00:00Z"
}
}
```

Submit Quiz Attempt:

Request:

POST /api/v1/quizzes/qiz_345678/take

Headers:

Authorization: Bearer <token>

Content-Type: application/json

Body:

```
{
  "answers": [
    { "questionId": "qst_901234", "selectedOption": 0 },
    { "questionId": "qst_901235", "selectedOption": 0 }
  ],
  "timeTaken": 45
}
```

Response (200 OK):

```
{
  "success": true,
  "data": {
    "attempt": {
      "id": "att_567890",
      "userId": "usr_123456",
      "quizId": "qiz_345678",
      "score": 2,
      "total": 2,
      "percentage": 100,
      "correctCount": 2,
      "incorrectCount": 0,
      "masteryLevel": "proficient",
      "timeTaken": 45,
      "completedAt": "2026-08-15T11:30:00Z",
      "results": [
        {
          "questionId": "qst_901234",
          "correct": true,
          "selectedOption": 0,
          "correctOption": 0,
          "explanation": "Water is composed of two hydrogen atoms and one oxygen atom."
        }
      ]
    }
  }
}
```

```
    },  
    {  
      "questionId": "qst_901235",  
      "correct": true,  
      "selectedOption": 0,  
      "correctOption": 0,  
      "explanation": "Carbon has 6 protons, so its atomic number is 6."  
    }  
  ]  
},  
"progressUpdate": {  
  "subjectId": "sub_789012",  
  "newMasteryLevel": 85,  
  "quizzesCompleted": 1  
}  
}  
}
```

Get progress overview:

Request:

GET /api/v1/progress/overview

Headers:

Authorization: Bearer <token>

Response (200 OK):

```
{  
  "success": true,  
  "data": {  
    "overview": {  
      "totalQuizzes": 25,  
      "averageScore": 72.5,  
      "studyStreak": 8,  

```

```
"lastStudyDate": "2026-08-15T09:00:00Z",
"subjects": [
  {
    "subjectId": "sub_789012",
    "name": "Chemistry",
    "icon": "🧪",
    "quizzesCompleted": 5,
    "averageScore": 82,
    "masteryLevel": 85
  },
  {
    "subjectId": "sub_789013",
    "name": "Mathematics",
    "icon": "📐",
    "quizzesCompleted": 8,
    "averageScore": 68,
    "masteryLevel": 70
  }
],
"areasForImprovement": [
  {
    "subject": "Mathematics",
    "topic": "Algebra",
    "score": 55,
    "recommendedQuiz": "Algebra Fundamentals"
  }
]
}
```

Database Schema

MongoDB Collections:

Users:

```
{
  _id: ObjectId,
  email: String (indexed, unique),
  passwordHash: String,
  name: String,
  avatar: String (URL),
  preferences: {
    language: String (default: "en"),
    theme: String (enum: ["light", "dark", "system"]),
    notifications: {
      push: Boolean,
      email: Boolean,
      sound: Boolean,
      vibration: Boolean
    }
  },
  progress: {
    totalQuizzes: Number,
    averageScore: Number,
    studyStreak: Number,
    lastStudyDate: Date
  },
  createdAt: Date,
  lastLogin: Date,
  isActive: Boolean,
  twoFactorEnabled: Boolean
}
```

Subject:

```
{
```

```
_id: ObjectId,
userId: ObjectId,
name: String,
icon: String,
color: String,
progress: {
  quizzesCompleted: Number,
  averageScore: Number,
  masteryLevel: Number (0-100)
},
createdAt: Date,
updatedAt: Date
}
Quizzes:
{
  _id: ObjectId,
  userId: ObjectId,
  subjectId: ObjectId,
  title: String,
  difficulty: String (enum: ["easy", "medium", "hard"]),
  questions: [{
    id: String,
    type: String (enum: ["multiple-choice", "true-false", "written"]),
    question: String,
    options: [String],
    correctAnswer: Number (index of correct option),
    explanation: String
  }],
  isPublic: Boolean,
  timesTaken: Number,
  averageScore: Number,
```

```
    createdAt: Date,
    updatedAt: Date
  }
  QuizAttempts:
  {
    _id: ObjectId,
    userId: ObjectId,
    quizId: ObjectId,
    answers: [{
      questionId: String,
      selectedOption: Number,
      isCorrect: Boolean
    }],
    score: Number,
    total: Number,
    percentage: Number,
    timeTaken: Number (seconds),
    completedAt: Date
  }
  Notifications:
  {
    _id: ObjectId,
    userId: ObjectId,
    type: String (enum: ["reminder", "result", "challenge", "system"]),
    title: String,
    body: String,
    data: Object (additional data),
    isRead: Boolean,
    createdAt: Date
  }
```

5.2 Request/Response Examples

Register User:

text

Request:

POST /api/v1/auth/register

```
{  
  "name": "John Doe",  
  "email": "john@example.com",  
  "password": "SecurePass123!",  
  "language": "en"  
}
```

Response (200):

```
{  
  "success": true,  
  "data": {  
    "user": {  
      "id": "usr_123456",  
      "name": "John Doe",  
      "email": "john@example.com",  
      "preferences": {  
        "language": "en",  
        "theme": "system",  
        "notifications": {  
          "push": true,  
          "email": true,  
          "sound": true,  
          "vibration": true  
        }  
      }  
    },  
    "createdAt": "2026-08-15T10:30:00Z"
```

```
},
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."
}
}
```

Create Quiz:

text

Request:

POST /api/v1/quizzes

```
{
  "title": "Chemistry Quiz 1",
  "subjectId": "sub_789012",
  "difficulty": "medium",
  "questions": [
    {
      "type": "multiple-choice",
      "question": "What is the chemical symbol for water?",
      "options": ["H2O", "CO2", "NaCl", "HCl"],
      "correctAnswer": 0,
      "explanation": "Water is composed of two hydrogen atoms and one oxygen atom."
    },
    {
      "type": "true-false",
      "question": "The atomic number of carbon is 6.",
      "options": ["True", "False"],
      "correctAnswer": 0,
      "explanation": "Carbon has 6 protons, so its atomic number is 6."
    }
  ]
}
```

Response (200):

```
{
  "success": true,
  "data": {
    "quizId": "qiz_345678",
    "title": "Chemistry Quiz 1",
    "subjectId": "sub_789012",
    "difficulty": "medium",
    "questions": [
      {
        "id": "qst_901234",
        "type": "multiple-choice",
        "question": "What is the chemical symbol for water?",
        "options": ["H2O", "CO2", "NaCl", "HCl"]
      },
      {
        "id": "qst_901235",
        "type": "true-false",
        "question": "The atomic number of carbon is 6.",
        "options": ["True", "False"]
      }
    ],
    "createdAt": "2026-08-15T11:00:00Z"
  }
}
```

Submit Quiz Attempt:

text

Request:

POST /api/v1/quizzes/qiz_345678/take

```
{
  "answers": [
    { "questionId": "qst_901234", "selectedOption": 0 },
  ]
}
```

```
{ "questionId": "qst_901235", "selectedOption": 0 }  
]  
}
```

Response (200):

```
{  
  "success": true,  
  "data": {  
    "attemptId": "att_567890",  
    "score": 2,  
    "total": 2,  
    "percentage": 100,  
    "correctCount": 2,  
    "incorrectCount": 0,  
    "masteryLevel": "proficient",  
    "results": [  
      {  
        "questionId": "qst_901234",  
        "correct": true,  
        "selectedOption": 0,  
        "correctOption": 0,  
        "explanation": "Water is composed of two hydrogen atoms and one oxygen atom."  
      },  
      {  
        "questionId": "qst_901235",  
        "correct": true,  
        "selectedOption": 0,  
        "correctOption": 0,  
        "explanation": "Carbon has 6 protons, so its atomic number is 6."  
      }  
    ]  
  }  
}
```

```
}
```

```
}
```

5.3 Database Schema

MongoDB Collections:

Users:

javascript

```
{
```

```
  _id: ObjectId,
```

```
  email: String (indexed, unique),
```

```
  passwordHash: String,
```

```
  name: String,
```

```
  avatar: String (URL),
```

```
  preferences: {
```

```
    language: String (default: "en"),
```

```
    theme: String (enum: ["light", "dark", "system"]),
```

```
    notifications: {
```

```
      push: Boolean,
```

```
      email: Boolean,
```

```
      sound: Boolean,
```

```
      vibration: Boolean
```

```
    }
```

```
  },
```

```
  progress: {
```

```
    totalQuizzes: Number,
```

```
    averageScore: Number,
```

```
    studyStreak: Number,
```

```
    lastStudyDate: Date
```

```
  },
```

```
  createdAt: Date,
```

```
  lastLogin: Date,
```

```
  isActive: Boolean,
```

twoFactorEnabled: Boolean

}

Subjects:

javascript

{

_id: ObjectId,

userId: ObjectId,

name: String,

icon: String,

color: String,

progress: {

quizzesCompleted: Number,

averageScore: Number,

masteryLevel: Number (0-100)

},

createdAt: Date,

updatedAt: Date

}

Quizzes:

javascript

{

_id: ObjectId,

userId: ObjectId,

subjectId: ObjectId,

title: String,

difficulty: String (enum: ["easy", "medium", "hard"]),

questions: [{

id: String,

type: String (enum: ["multiple-choice", "true-false", "written"]),

question: String,

options: [String],

```
    correctAnswer: Number (index of correct option),
    explanation: String
  }},
  isPublic: Boolean,
  timesTaken: Number,
  averageScore: Number,
  createdAt: Date,
  updatedAt: Date
}
```

QuizAttempts:

```
javascript
{
  _id: ObjectId,
  userId: ObjectId,
  quizId: ObjectId,
  answers: [{
    questionId: String,
    selectedOption: Number,
    isCorrect: Boolean
  }],
  score: Number,
  total: Number,
  percentage: Number,
  timeTaken: Number (seconds),
  completedAt: Date
}
```

Notifications:

```
javascript
{
  _id: ObjectId,
  userId: ObjectId,
```

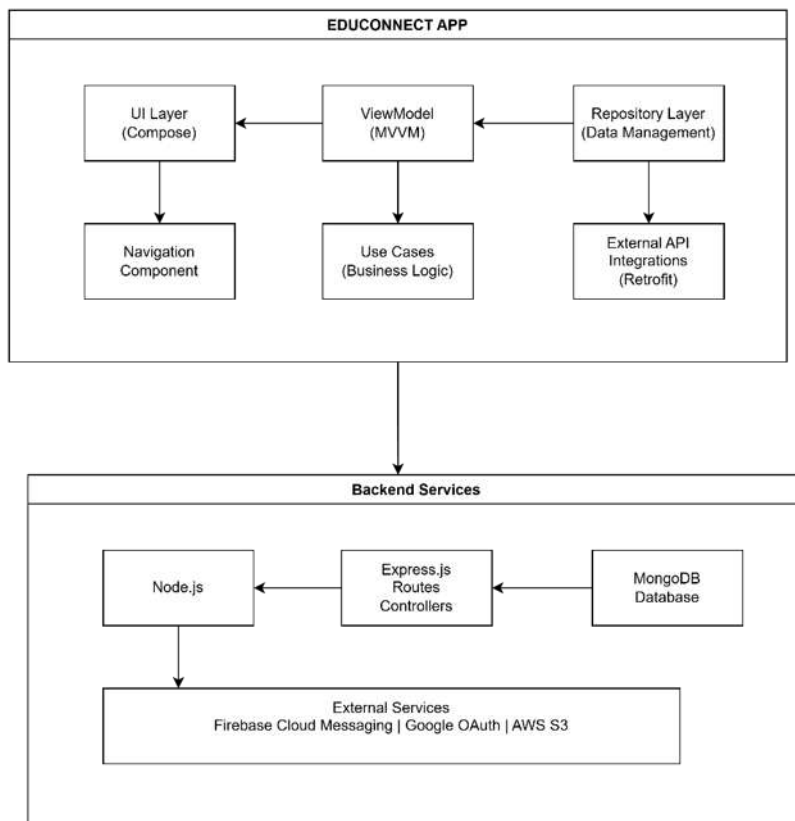
```

type: String (enum: ["reminder", "result", "challenge", "system"]),
title: String,
body: String,
data: Object (additional data),
isRead: Boolean,
createdAt: Date
}

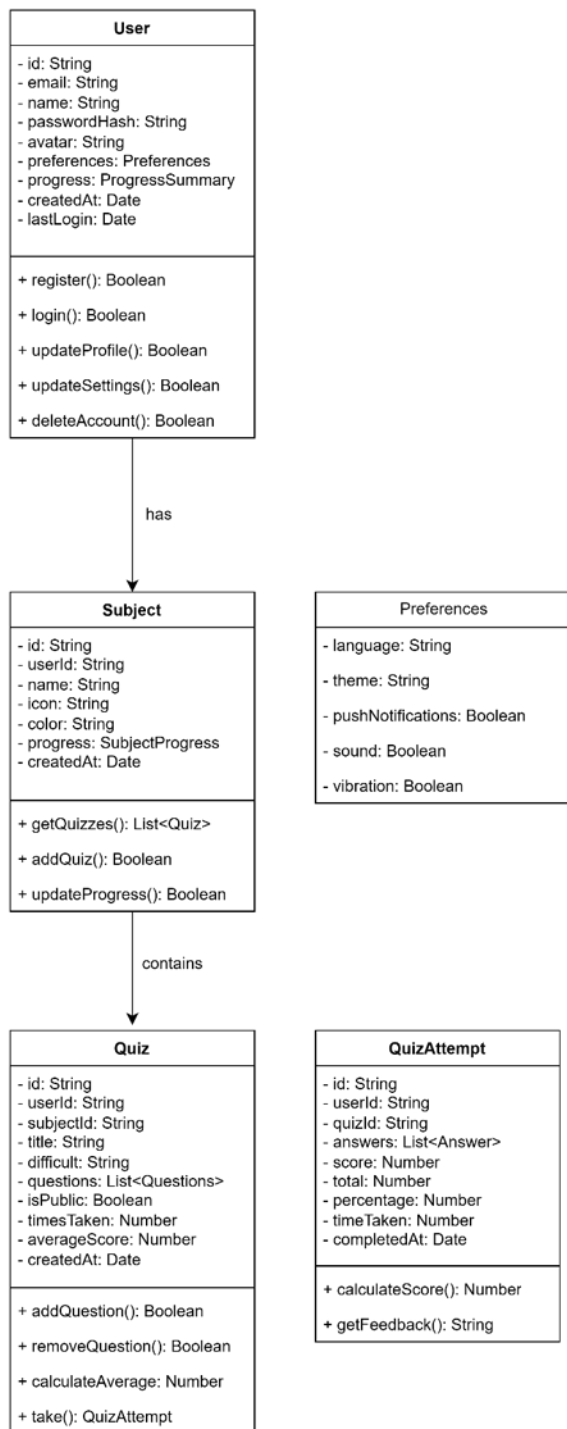
```

6. UML Diagrams

6.1 System Architecture Diagram



6.2 Class Diagram (Core Components)



7. Data Requirements

7.1 Data Types and Storage

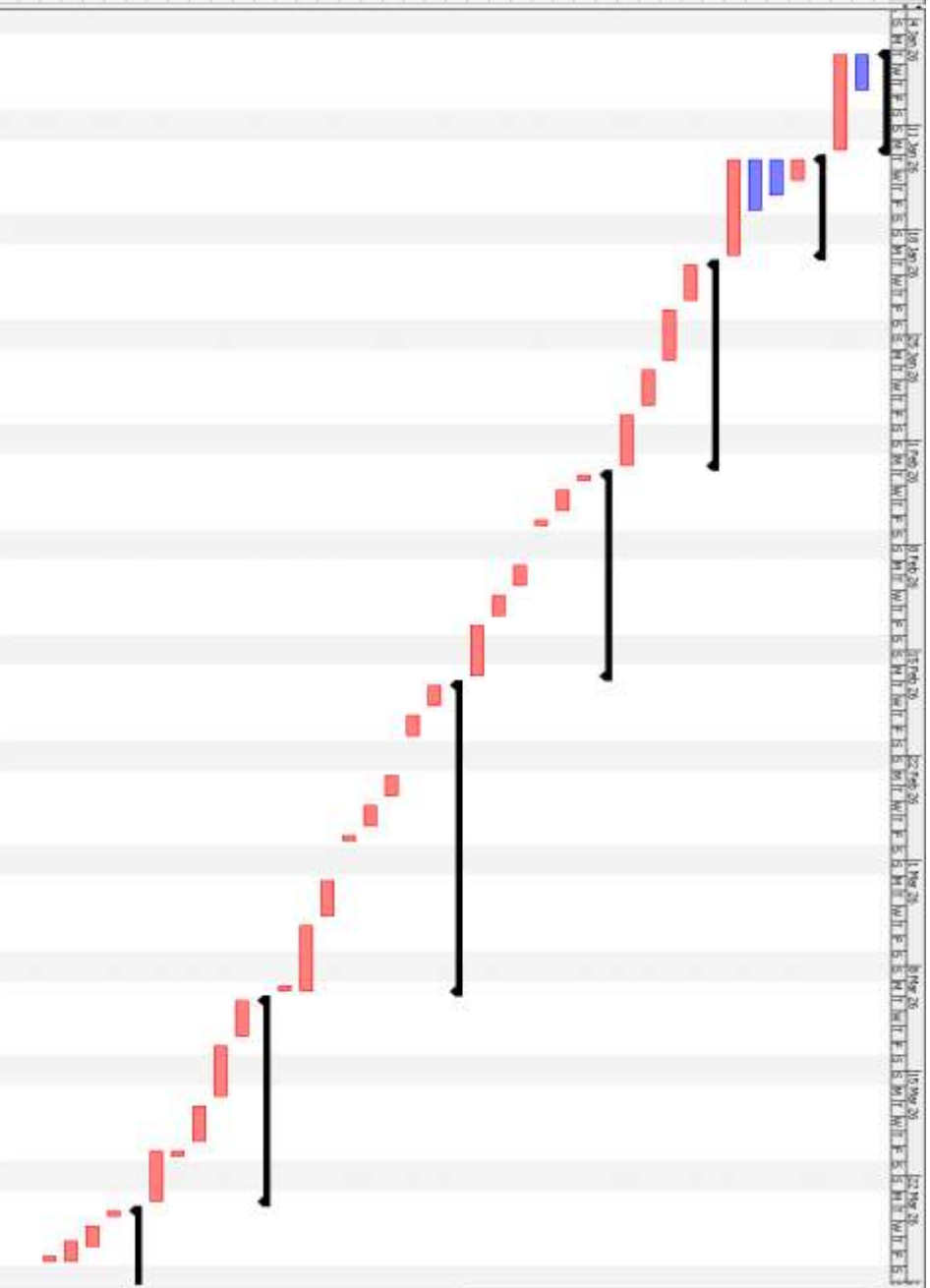
Data Entity	Fields	Data Type	Storage Location
User	userId, name, email, password, role, language	String, String, String, String, String, String	Firebase Auth + Firestore
Course	courseId, name, teacher, description	String, String, String, String	Firestore
Assignment	assignmentId, title, description, dueDate, courseId	String, String, String, Date, String	Firestore
Grade	gradeId, assignmentId, studentId, score, total	String, String, String, Double, Double	Firestore
Announcement	announcementId, title, content, date	String, String, String, Date	Firestore
Notification	notificationId, userId, title, message, read, date	String, String, String, Boolean, Date	Firestore
OfflineData	id, data, timestamp	String, JSON, Date	Room Database (SQLite)

7.2 Data Capture Requirements

Screen	Data Captured	Data Type	Validation
Register	Name, Email, Password, Confirm Password	String, String, String, String	Email format, Password minimum 6 chars
Login	Email, Password	String, String	Email format, Not empty
Settings	Name, Email, Language	String, String, String	Email format
Change Password	Old Password, New Password, Confirm Password	String, String, String	New password minimum 6 chars
Course Detail	None	-	-
Assignments	None	-	-
Grades	None	-	-

8. Project Plan (Gantt Chart)

	📌	Name	Duration	Start	Finish
1	🟢	E1. Research Phase	3 days	2026/01/06, 08:00	2026/01/12, 17:00
2	🟢	Competitor Analysis	3 days	2026/01/06, 08:00	2026/01/08, 17:00
3	🟢	Technology Research	5 days	2026/01/06, 08:00	2026/01/12, 17:00
4	🟢	E2. Planning Phase	5 days	2026/01/13, 08:00	2026/01/19, 17:00
5	🟢	Requirements Definition	2 days	2026/01/13, 08:00	2026/01/14, 17:00
6	🟢	Functional Requirements	3 days	2026/01/13, 08:00	2026/01/15, 17:00
7	🟢	Non-Functional Requirements	4 days	2026/01/13, 08:00	2026/01/16, 17:00
8	🟢	System Architecture	5 days	2026/01/13, 08:00	2026/01/19, 17:00
9	🟢	E3. Design Phase	10 days	2026/01/20, 08:00	2026/02/02, 17:00
10	🟢	UI/UX Wireframes	3 days	2026/01/20, 08:00	2026/01/22, 17:00
11	🟢	User Interface Design	2 days	2026/01/23, 08:00	2026/01/25, 17:00
12	🟢	API Design	3 days	2026/01/27, 08:00	2026/01/29, 17:00
13	🟢	Database Design	2 days	2026/01/20, 08:00	2026/01/22, 17:00
14	🟢	E4. Backend Development	10 days	2026/02/03, 08:00	2026/02/16, 17:00
15	🟢	Project API Setup	1 day	2026/02/03, 08:00	2026/02/03, 17:00
16	🟢	User Authentication	2 days	2026/02/04, 08:00	2026/02/05, 17:00
17	🟢	User Management	1 day	2026/02/06, 08:00	2026/02/06, 17:00
18	🟢	Post Management API	2 days	2026/02/07, 08:00	2026/02/10, 17:00
19	🟢	Social Media API Integration	2 days	2026/02/11, 08:00	2026/02/12, 17:00
20	🟢	Content Implementation	2 days	2026/02/13, 08:00	2026/02/16, 17:00
21	🟢	E5. Frontend Development	13 days	2026/02/17, 08:00	2026/03/04, 17:00
22	🟢	Analytics/Featured Project 5	2 days	2026/02/17, 08:00	2026/02/18, 17:00
23	🟢	Logout/Feedback Screens	2 days	2026/02/19, 08:00	2026/02/20, 17:00
24	🟢	Dashboard UI	2 days	2026/02/23, 08:00	2026/02/24, 17:00
25	🟢	Post Creation Screen	2 days	2026/02/26, 08:00	2026/02/28, 17:00
26	🟢	Post Scheduling	1 day	2026/02/27, 08:00	2026/02/27, 17:00
27	🟢	API Integration	3 days	2026/03/02, 08:00	2026/03/04, 17:00
28	🟢	Content Placeholder	3 days	2026/03/05, 08:00	2026/03/09, 17:00
29	🟢	UI Refinement	1 day	2026/03/09, 08:00	2026/03/09, 17:00
30	🟢	E6. Testing Phase	10 days	2026/03/16, 08:00	2026/03/23, 17:00
31	🟢	User Testing	3 days	2026/03/16, 08:00	2026/03/18, 17:00
32	🟢	UI Testing	2 days	2026/03/17, 08:00	2026/03/16, 17:00
33	🟢	API Integration Testing	3 days	2026/03/17, 08:00	2026/03/19, 17:00
34	🟢	Bug Identification	1 day	2026/03/20, 08:00	2026/03/20, 17:00
35	🟢	Bug Fixing & Retesting	2 days	2026/03/23, 08:00	2026/03/23, 17:00
36	🟢	E7. Deployment	5 days	2026/03/24, 08:00	2026/03/30, 17:00
37	🟢	Production Configuration	1 day	2026/03/24, 08:00	2026/03/24, 17:00
38	🟢	Final Build	2 days	2026/03/25, 08:00	2026/03/26, 17:00
39	🟢	Deployment	2 days	2026/03/26, 08:00	2026/03/27, 17:00
40	🟢	Final Verification	1 day	2026/03/27, 08:00	2026/03/27, 17:00
41	🟢	Documentation & Handoff	1 day	2026/03/28, 08:00	2026/03/28, 17:00



(ProjectLibre, 2026)

Key Milestones

Milestone	Completion Date
Milestone 1: Research Complete	August 17
Milestone 2: Planning Complete	August 24
Milestone 3: Design Complete	August 31
Milestone 4: Backend Development Complete	September 14
Milestone 5: Frontend Development Complete	September 30
Milestone 6: Testing & Bug Fixing Complete	October 24
Milestone 7: Deployment Ready	November 7

What this section demonstrates

This satisfies the requirement because you are showing:

- What tasks must be completed
- Smaller subtasks for each major activity
- How long each task will take
- When each task starts and ends
- Testing time
- Bug-fixing and retesting time
- Final deployment
- Important project milestones

The most important improvement over simply writing “Implement Prototype 20 days” is that you've broken implementation into things such as authentication, user management, post creation, scheduling, API integration, offline mode, UI refinement, testing, bug identification, and bug fixing. That makes the project plan much more realistic and demonstrates that you understand the development process.

Task Breakdown

The development of EduConnect will be divided into several stages. Each stage focuses on a specific part of the project to ensure that the application is properly researched, planned, designed, developed, tested, and prepared for release.

Research Phase (Aug 17–21)

The first stage will focus on understanding existing study applications and deciding which technologies are most suitable for EduConnect.

- **Competitor Analysis:** Research popular learning platforms such as Quizlet, Kahoot!, and Socratic by Google. The goal is to identify their strengths, weaknesses, and useful features that could inspire EduConnect.
- **Technology Research:** Investigate different Android development technologies, APIs, and database options to determine which tools will work best for the application.

Planning Phase (Aug 24–28)

During this stage, the project requirements and overall system structure will be established.

- **Requirements Definition:** Finalise the functional and non-functional requirements of EduConnect, including what the application must do and the standards it should meet.
- **Architecture Design:** Plan the overall structure of the application and select the most suitable technology stack for development.

Design Phase (Aug 31–Sep 11)

The design stage will focus on creating the visual appearance and technical structure of the application.

- **UI/UX Design:** Develop wireframes, screen mockups, and navigation flows to ensure that the application is easy and intuitive to use.
- **API Design:** Plan the RESTful API, including the different endpoints and the information that will be sent and received between the application and server.
- **Database Design:** Create the MongoDB database structure and determine how information such as users, subjects, quizzes, and progress will be stored.

Backend Development (Sep 14–29)

The backend development stage will focus on creating the server-side functionality required by EduConnect.

- **API Development:** Develop the API endpoints needed for authentication, user management, subjects, quizzes, and student progress.
- **Database Implementation:** Set up the MongoDB collections, relationships, and indexes required to store and retrieve application data efficiently.

Frontend Development (Sep 30–Oct 23)

This stage focuses on building the Android application and connecting it to the backend services.

- **Android Setup:** Configure the Android Studio project and install the required libraries and dependencies.
- **UI Implementation:** Build the application's screens and user interface using Jetpack Compose.
- **API Integration:** Connect the Android application to the backend using Retrofit so that users can send and retrieve data.

- **Offline Mode:** Implement RoomDB to store information locally and allow students to continue using the application when they do not have an internet connection. Data will be synchronised when the connection becomes available again.

Testing Phase (Oct 24–Nov 6)

Once the main development work is complete, the application will undergo thorough testing to identify and resolve problems.

- **Unit Testing:** Create JUnit tests to check important components such as ViewModels and Repositories.
- **Integration Testing:** Test the communication between the Android application, API, and database to make sure information is transferred correctly.
- **Bug Fixing:** Identify and fix errors while also improving the application's performance, reliability, and overall user experience.

Deployment (Nov 7–13)

The final stage will prepare EduConnect for release.

- **Google Play Store Preparation:** Generate the required APK or AAB file and prepare the application's store listing, screenshots, description, and other required information.
- **Production Deployment:** Publish the completed application to the Google Play Store once all final checks have been completed.

9. Conclusion

EduConnect is designed as a comprehensive learning and quiz application that aims to address some of the challenges experienced by South African students. By combining useful features found in existing educational platforms with new functionality and an inclusive design, the application aims to make studying more personalised, interactive, and accessible.

The proposed application includes important features such as secure user authentication, encrypted data, Single Sign-On (SSO), account and settings management, REST API integration, offline learning with synchronisation, real-time notifications, and support for English, isiZulu, and Afrikaans. The development process will also follow modern Android development practices, including the MVVM architecture, separation of responsibilities between different parts of the system, and structured testing.

The project schedule provides a clear development path, with each phase having specific tasks and deadlines. Dividing the project into research, planning, design, development, testing, and deployment makes it easier to monitor progress and manage the workload. Dedicated time for testing and fixing bugs will also help ensure that the final application is stable and meets the required academic and user expectations.

Overall, EduConnect has the potential to be a useful educational tool for students in South Africa. Its combination of simplicity, accessibility, personalisation, and interactive learning features can support students with different learning needs and levels of digital access. With careful development and testing, the application can provide students with a practical way to study, monitor their progress, and work towards achieving their academic goals.

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